

Intelligent Farm Operations Platform with Smart Agro Precision System.

Farming : Crop Recommendations, Disease Detection, Smart Irrigation, Satellite & Drone Monitoring and Farmer Analytics.

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⚠️ Problem Statement

Challenges Faced by Farmers

1 No Real-Time Monitoring

Manual observation.
Delayed detection of crop stress.

2 Resource Wastage

Over-irrigation wastes 40-60% water.
Excess fertilizer degrades soil.

3 Unpredictable Weather

No early warning systems.
Crop failure from sudden changes.

4 Late Pest Detection

Identified too late.
₹90,000 Cr+ annual crop loss.

5 No Expert Access

Small farmers lack guidance.
Language barriers.

6 Fragmented Data

No centralized platform.
Decisions based on intuition.

💡 OBJECTIVES-FARMEVERSE

- FarmVerse is a smart farming platform designed to support farmers in managing their farms more effectively.
- It helps farmers make informed decisions to improve crop growth and productivity.
- It promotes the efficient use of resources such as water, fertilizers, and land
- It assists in monitoring farm conditions and responding to changes in weather.
- It helps in detecting the disease early.
- Improves overall farm productivity while reducing operational cost.
- Support timely decisions for better farming outcomes.
- Encourage sustainable and environmentally friendly farming practices.
- Increase farmers' profitability by optimizing the use of available resources





FarmVerse

Precision Agriculture Management Platform — Modules



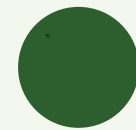
Farmer Register Module

farmer create an account



Farm Management Module

Add & manage fields, crops and daily farm activities



Sensor Monitoring Module

Live soil moisture, temperature , humidity ,soil pH, High Intensity, Water level



Weather Monitoring Module

Real-time forecasts, rainfall & climate risk updates



Smart Irrigation Module

scheduling to save water ,Prevents over-irrigation,reduce electricity cost



Crop Recommendation Module

based on soli pH,temparature,humidity,rainfall



Disease Detection Module

Image-based detection of crop diseases at early stage



Fertilizer Recommendation Module

Suggests right fertilizer type & dosage for each crop



Alert Module

Instant SMS/app alerts for risks, weather & sensor events



Dashboard Module

Unified view of farm health, analytics & reports

Unique Features

How FarmVerse Works Differently



Smart Dashboard

Map-based field viz
Real-time sensor overlay
Zone your farm into parcels

✦✦ Geospatial + temporal fusion



AI Crop Advisory

Upload leaf → instant detection
50K+ disease images trained
Works with smartphone

✦✦ Camera-based, no hardware



Smart Irrigation

Soil moisture + weather API
AI calculates exact water
Automated scheduling

✦✦ Saves 30-40% water

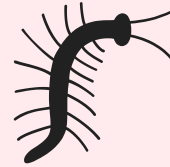
More Unique Features

Setting FarmVerse Apart



Weather

Micro-level predictions
Field-specific alerts
Frost/heatwave/rain warnings



Pest Prediction

7-14 day advance alerts
Weather + historical data
Preventive approach



Yield Prediction

ML-based forecasting
Historical trends
Market strategy planning



Marketplace

Buy seeds/fertilizers
Sell produce directly
Fair pricing



Voice Assistant

Local languages
Voice-based advisory
For non-tech farmers



Offline-First

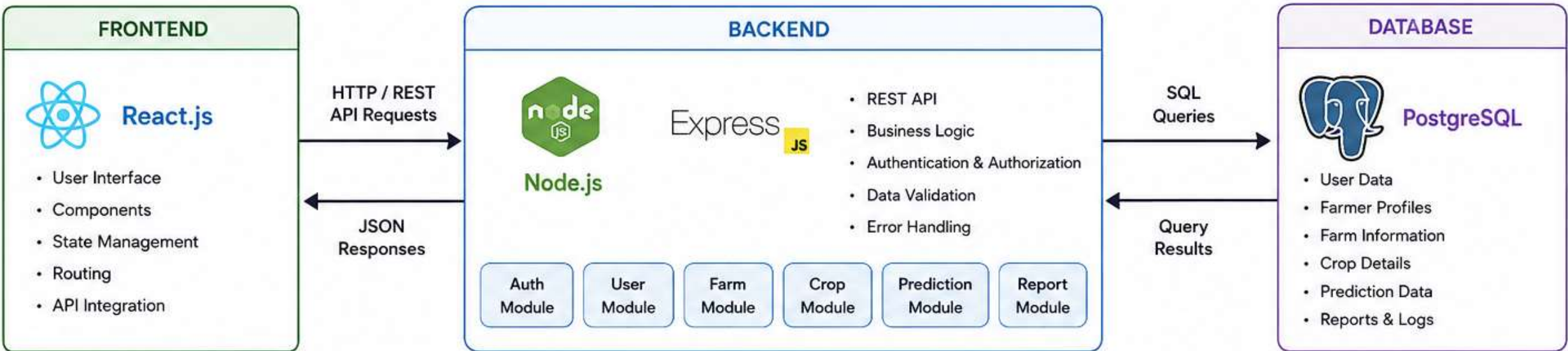
Works without internet
Syncs when connected
Rural India ready

? System Architecture

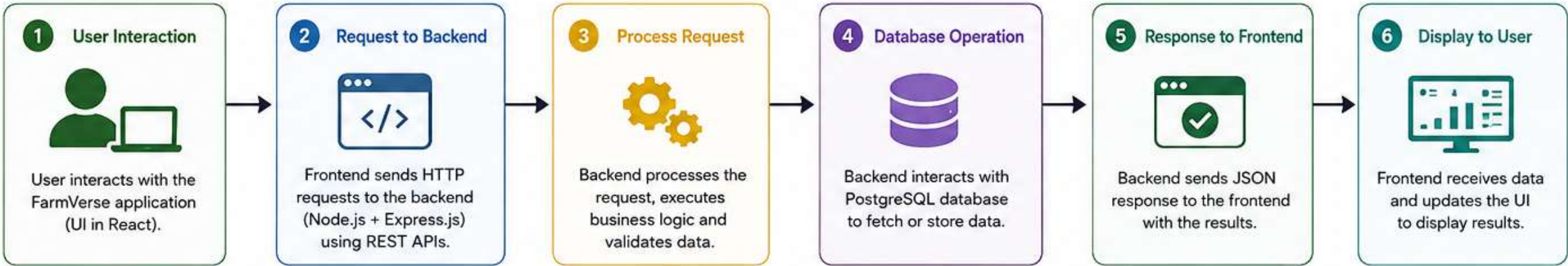
Smart, AI-Powered & Connected Farming Platform

FarmVerse – System Architecture

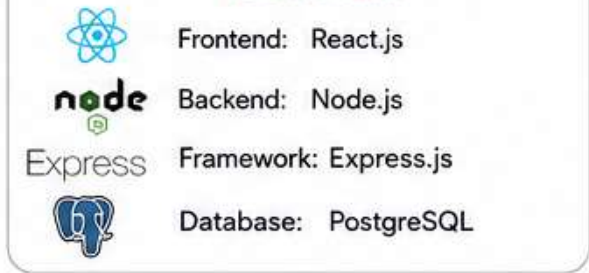
Smart, AI-Powered Agriculture Management Platform



WORKFLOW



TECH STACK



KEY FEATURES HANDLED



Better Farming • Higher Yields • Sustainable Future

Technology Stack

Tools & Technologies

FRONTEND

 React.js

 Tailwind CSS

BACKEND

 Node.js + Express

 Python FastAPI

 GraphQL + REST

 JWT Auth

DATABASE

 PostgreSQL

CLOUD & TOOLS

 GitHub Actions

 Vs code , IntelliJ IDEA

EXTERNAL APIs

 OpenWeatherMap

Key Features

Platform Capabilities

 Smart Map

 AI Advisor

 Irrigation

 Weather

 Pest Control

 Analytics

 Marketplace

 Voice Bot

 Community

 Mobile App

 Alerts

 Reports

✓ Benefits

Impact & Advantages



FOR FARMERS

📈 20-30% yield increase

💧 30-40% water savings

📦 25% less chemicals

💰 Better market prices



FOR ENVIRONMENT

📦 Reduced chemical runoff

💧 Water conservation

🌱 Sustainable farming

📦 Lower carbon footprint



FOR ECONOMY

📈 Higher profitability

🍏 Reduced food waste

🌐 Rural digitization

👤💻 Agri-tech jobs

Future Scope



Satellite Imagery

NDVI crop health from space
Large-scale monitoring
Detect stress before visible



Drone Monitoring

Automated aerial surveys
Precision spraying
3D terrain mapping



Advanced Analytics

ROI calculators
Carbon credit tracking
Water footprint analysis

Future Scope



Govt Integration

PM-KISAN link
Crop insurance
Subsidy recommendations



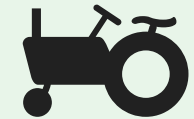
IoT Expansion

Smart greenhouses
Livestock monitoring
Aquaculture integration



Crop Genome DB

AI-recommended varieties
Soil-crop matching
Hybrid seed suggestions



Autonomous Farming

Robot harvesting
Self-driving tractors
Automated planting

CONCLUSION -

Transforming Farming with Smart Technology

- FarmVerse provides an integrated platform for smarter and more efficient farm management.
- Combines AI, IoT, weather data, sensors, and analytics to support timely farming decisions.
- Features such as crop recommendation, disease detection, smart irrigation, weather monitoring, and farmer analytics help optimize resources.
- The platform aims to reduce water and chemical usage, improve crop productivity, and increase farmer profitability.
- With future integration of satellite imagery, drones, advanced analytics, IoT, and autonomous farming, FarmVerse can support scalable and sustainable agriculture.

“Smarter Decisions • Better Yields • Sustainable Farming”

Thank You